MTH203: Sequences and series
Lecture 06

1. Sequences

Recall that we had introduced sequences in Lecture 1. We had not discussed functions at that point and so our definition was somewhat informal. We will now define a *sequence* in a set S to be a function $f : \mathbb{N} \to S$. We will use slightly different notation for these functions. The element $f(n)$ will be represented by f_n and so we will say that $(f_1, f_2, \ldots,)$ is a sequence.

More generally, if k is any integer, let $I_k = \{k, k+1, \ldots\} = \{n \in \mathbb{Z} | n \geqslant k\}$. Then, we may define a sequence in S to be a function $f : I_k \to S$ and denote it by $(f_k, f_{k+1}, \ldots)$. This does not give rise to a different concept since we have the obvious bijection $\mathbb{N} \to I_k$ by mapping n to $(n + k - 1)$.

2. Decimal representations

We will now formally describe the familiar notion of *decimal representation* of a real number.

Let x be a real number, $x \geqslant 0$. We will now inductively construct a sequence of rational numbers which are meant to serve as approximations for x.

Recall that any bounded above subset of the integers has a maximal element (see Lecture 5, Theorem 3). We consider the set $\{n \in \mathbb{N} \cup \{0\} | n \leqslant x\}$. This set has x has an upper bound, and so it has a maximal element. We denote this element by a_0. This is the largest integer that is less than or equal to x.

More generally, for any integer $k \in \mathbb{N} \cup \{0\}$, we define a_k to be the maximal element of the set $\{n \in \mathbb{N} \cup \{0\} | n/10^k \leqslant x\}$. Thus, a_k is the largest integer such that $a_k/10^k \leqslant x$.

For any non-negative integer k, we define $x_k := a_k/10^k$. By definition, for any non-negative integer k, we have $(a_k + 1)/10^k > x$. Thus, $x_k = a_k/10^k$ is the largest rational number that can be written in the form $m/10^k$ which is less than or equal to x.

As we saw above, for any non-negative integer k, $(a_k + 1)/10^k > x$, and so $x - x_k = x - (a_k/10^k) < 1/10^k$. Thus, for every k, we have the inequalities

$$0 \leqslant x - x_k \leqslant 1/10^k.$$

The numbers in the sequence $1/10, 1/10^2, \ldots$ become as small as well like. In other words, we have the following:

Lemma 1. *Let $\epsilon > 0$ be any real number. Then, there exists a non-negative integer k such that $1/10^k < \epsilon$.*

Proof. We have $1/10^k < \epsilon$ if and only if $10^k > 1/\epsilon$. Thus, it suffices to show that there exists a non-negative integer k such that $10^k > 1/\epsilon$.

Suppose that $10^k < 1/\epsilon$ for all integers k. Then the set $T = \{10^k\}_{k=0}^{\infty}$ is bounded above by $1/\epsilon$ and has a least upper bound. Let $\alpha = \sup(T)$.

Since $\alpha - 1$ is less than $\sup(T)$, it is not an upper bound for T. Thus, there exists an integer k such that $\alpha - 1 < 10^k$. Multiplying this inequality by 10 (which is a positive number), we see that $10(\alpha - 1) < 10^{k+1}$. But as $10^{k+1} \in T$, we have $10^{k+1} < \alpha$. Thus, $10(\alpha - 1) < \alpha$. Rearranging this inequality, we get $10\alpha - \alpha < 10$, i.e. $9\alpha < 10$. Thus, $\alpha < 10/9 < 10$, which is a contradiction. Thus, T cannot be bounded above. This completes the proof. □

Since $x \geqslant x_k$, we see that x is an upper bound for the set $S := \{x_k\}_{k=0}^{\infty}$. Given any real number $\epsilon > 0$, the above lemma shows that there exists a non-negative integer k such that $1/10^k < \epsilon$. Thus, $x - x_k < 1/10^k < \epsilon$. By the characterization for supremums obtained in Lemma 7 of Lecture 4, we conclude that x is the supremum of the set $\{x_0, x_1, \ldots\}$.

For every k, we have $10a_k/10^{k+1} = a_k/10^k \leqslant x$. Thus, by the definition of a_{k+1}, it follows that $10a_k < a_{k+1}$. Also, since $10(a_k+1)/10^k = (a_k+1)/10^k > x \geqslant a_{k+1}/10^{k+1}$, we see that $10(a_k+1) > a_{k+1}$. Thus, $a_{k+1} - 10a_k < 10$.

Thus, we have obtained the inequalities $0 \leqslant a_{k+1} - 10a_k < 10$. Thus, $a_{k+1} - 10a_k$ is in the set $\{0, 1, 2, \ldots, 9\}$. For every integer $k \geqslant 1$, we define $d_k := a_k - a_{k-1}$. This is called as the *k-th digit after the decimal point in the decimal representation of x.* We say that the expression

$$a_0.d_1d_2d_3\ldots d_n\ldots$$

is the *decimal representation* of x. We will, in fact, write the following equality:

$$x = a_0.d_1d_2d_3\ldots d_n\ldots$$

Note that, this is a *definition.* The expression on the right is to be interpreted as a symbol for the real number on the left hand side.

Example 2. Suppose $x = 1/7$. Then, $x_0 = 0$ since $0 \leqslant x < 1$.

Since $1/10 < 1/7$, but $2/10 > 1/7$ (which can be verified from the fact that $2 \times 7 = 14 > 1 \times 10$), we see that $a_1 = 1$. Thus $x_1 = 1/10$.

Computing in this manner, we can verify the following:

$a_2 = 14$ and so $x_2 = 14/100$
$a_3 = 142$ and so $x_3 = 142/1000$
$a_4 = 1428$ and so $x_4 = 1428/10000$

... ... and so on. Thus, $1/7 = 0.142857142857\ldots$, which is the familiar decimal representation of $1/7$.

Thus, every non-negative real number has a decimal representation. If x is a negative real number, and if $a_0.d_1d_2\ldots$ is the decimal representation of $-x$, then we say that

$$x = -a_0.d_1d_2\ldots$$

is the decimal representation of x. Thus, we have a decimal representation for every real number.

However, given any a_0 is a non-negative integer and a sequence $(d_i)_{i=1}^{\infty}$ for $0 \leqslant d_i \leqslant 9$ for all i, does the expression $a_0.d_1d_2\ldots$ correspond to some real number? Let us explore this question.

3. Numbers with a given decimal representation

Suppose that we are given any non-negative integer a_0 and any sequence $(d_1, d_2, \ldots)$ of integers d_i satisfying $0 \leqslant d_k \leqslant 9$. We define the sequence $(x_0, x_1, \ldots)$ of real numbers by setting $x_0 = a_0$ and $x_k = a_0 + \sum_{i=0}^{k} d_i/10^i$ for any $k \geqslant 1$. Then, we have $x_k \leqslant x_{k+1}$ for any non-negative integer k. We claim that the set $\{x_0, x_1, \ldots\}$ is bounded above by $x_0 + 1$. This is proved as follows:

Lemma 3. *For any positive integer k, $x_k \leqslant x_0 + 1$.*

Proof. We will use induction on k to prove the stronger claim that $(x_0+1) - x_k \geqslant 1/10^k$.

For $k = 0$, we have $(x_0+1) - x_0 = 1 \geqslant 1/10^0$. Thus, the claim is true in this case. Let us now assume that for some integer $k \geqslant 0$, we have $(x_0+1) - x_k \geqslant 1/10^k$.

Recall that $x_{k+1} = x_k + d_{k+1}/10^{k+1}$. Also, recall that $d_{k+1} \leqslant 9$. Using these facts, we have

$$\begin{aligned}(x_0+1) - x_{k+1} &= (x_0+1) - \Big(x_k + d_{k+1}/10^{k+1}\Big) \\ &= [(x_0+1) - x_k] - d_{k+1}/10^{k+1} \\ &\geqslant 1/10^k - d_{k+1}/10^{k+1} \\ &\geqslant 1/10^k - 9/10^{k+1} \\ &= 1/10^{k+1}.\end{aligned}$$

This completes the induction step and hence completes the proof. □

Let x be the supremum of the set $\{x_0, x_1, \ldots\}$. It is tempting to think that the decimal representation of x must be $a_0.d_1d_2\ldots$, but this is not always the case!

Here is an example:

Example 4. Let $a_0 = 0$ and let $d_k = 9$ for all positive integers k. We set $x_k = a_0 + \sum_{i=0}^{k} d_i/10^i$ and let $x = \sup\{x_k\}_{k=0}^{\infty}$. We claim that $x = 1$.

Indeed, one can easily verify (by induction on k), that $1 - x_k = 1/10^k$. (*Exercise:* Write out the detailed proof of this fact.) Thus, 1 is an upper bound for the set $\{x_k\}_{k=0}^{\infty}$. If ϵ is any positive real number, we choose (using Lemma 1) a positive integer k such that $1/10^k < \epsilon$. Then,

$$x - x_k = 1/10^k < \epsilon.$$

Thus, $x - \epsilon < x_k$ and so $x - \epsilon$ is not an upper bound of $\{x_k\}_{k=0}^{\infty}$. Thus, x is the least upper bound, i.e. the supremum of $\{x_k\}_{k=0}^{\infty}$.

Thus, it seems that $0.999\ldots$ should be the decimal representation of 1. But then our procedure for computing the decimal representation actually gives us $1.000\ldots$ as the decimal representation of 1. Thus, not every expression of the form $a_0.d_1d_2\ldots$ is the decimal representation of some non-negative real number (according to our definition of decimal representation).

It turns out that this is essentially the only scenario where things go wrong. The thing that is causing the problem is the infinite sequence of 9's. Let us see why this is so. The following argument is a little tricky to think of, but studying it is worthwhile because it will help you understand the concepts we have introduced so far.

For every non-negative integer k, let us define $y_k := x_k + 1/10^k$.

Lemma 5. *For every non-negative integer k, we have $y_k \geqslant y_{k+1}$. Equality holds if and only if $d_{k+1} = 9$.*

Proof. This is proved by the computation

$$\begin{aligned}y_k - y_{k+1} &= x_k + 1/10^k - x_{k+1} - 1/10^{k+1} \\ &= x_k + 1/10^k - (x_k + d_{k+1}/10^{k+1}) - 1/10^{k+1} \\ &= \frac{10 - d_{k+1} - 1}{10^k} \\ &\geqslant 0,\end{aligned}$$

where equality holds if and only if $d_{k+1} = 9$. □

Thus, $y_1 \geqslant y_2 \geqslant y_3 \geqslant \ldots$. In other words, the sequence $(y_k)_{k=1}^{\infty}$ is a non-increasing sequence. (Recall that the sequence $(x_k)_{k=0}^{\infty}$ is a non-decreasing sequence since $x_{k+1} = x_k + d_{k+1}/10^{k+1} \geqslant x_k$ for every non-negative integer k.)

Lemma 6. *For any non-negative integers, k and l, we have $x_k < y_l$.*

Proof. If $k \leqslant l$, then $x_k \leqslant x_l < x_l + 1/10^l = y_l$. On the other hand, if $k > l$, then $x_k < x_k + 1/10^k = y_k \leqslant y_l$. □

This gives us a very interesting relationship between these two sequences:

Lemma 7. $\sup\{x_k\}_{k=0}^{\infty} = x = \inf\{y_k\}_{k=0}^{\infty}$.

Proof. Here, the first equality is just the definition of x. We need to prove the second equality.

Let us fix a non-negative integer l. Then, as we saw in the previous lemma, $x_k \leqslant y_l$ for all non-negative integers k. Thus, y_l is an upper bound for the set $\{x_k\}_{k=0}^{\infty}$. Since x is the supremum of this set, we have $x \leqslant y_l$.

Thus, x is a lower bound for the set $\{y_l\}_{l=0}^{\infty}$. We want to show that it is the greatest lower bound. So, let $\epsilon > 0$ be any real number. We will show that $x + \epsilon$ is not a lower bound for the set $\{y_l\}_{l=0}^{\infty}$.

Using Lemma 1, we choose an integer l such that $1/10^l < \epsilon$. Then, $y_l = x_k + 1/10^l \leqslant x + \epsilon$. Thus, we see that $x + \epsilon$ is not a lower bound for $\{y_l\}_{l=0}^{\infty}$.

Thus, we see that x is the greatest lower bound, i.e. infimum of $\{y_l\}_{l=0}^{\infty}$. □

Recall that we are trying to investigate the following question:

Suppose we are given a non-negative integer a_0 and a sequence $(d_k)_{k=1}^{\infty}$ of integers satisfying $0 \leqslant d_k \leqslant 9$ for all k. Then, if we define $x_k = a_0 + \sum_{i=1}^{k} d_i/10^i$ for all k, and set x to be the supremum of the set $\{x_k\}_{k=1}^{\infty}$, then is the decimal representation of x equal to $a_0.d_1d_2\ldots$?

We are now in a position to answer this question. What do we need for $a_0.d_1d_2\ldots$ to be the decimal representation of x? Suppose we define $a_k := 10x_k$, then we would see if a_k is the largest integer in the set

$$S_k := \{n \in \mathbb{N} \cup \{0\} | n/10^k \leqslant x\}.$$

By construction, $a_k/10^k = x_k \leqslant x$, and so a_k is an element of S_k. We would like to check whether $a_k + 1$ is in S_k.

The integer $(a_k + 1)$ will fail to be in S_k if and only if we have the inequality

$$x < (a_k + 1)/10^k = x_k + 1/10^k = y_k.$$

This will happen if and only if there is some $l > k$ such that $y_l < y_k$. This happens if and only if $d_l < 9$. Thus, we see that a_k is the largest element of the set S_k if and only if there is some $l > k$ such that $d_l < 9$.

Thus, we conclude the following:

Theorem 8. *Let a_0 be a non-negative integer and let $(d_i)_{i=1}^{\infty}$ be a sequence of non-negative integers such that the following conditions hold:*

(a) $0 \leqslant d_i \leqslant 9$ for all natural numbers i.
(b) For every natural number k, there exists a natural number l such that $k < l$ and $d_l < 9$.

Then, if we define $x_k = a_0 + \sum_{i=1}^{k} d_i/10^i$ for all k, the set $\{x_k\}_{k=0}^{\infty}$ is bounded above and the decimal representation of its supremum is given by

$$a_0.d_1d_2\ldots.$$

Condition (b) in the statement of the above theorem just says that decimal representations of non-negative real numbers do not end with an infinite sequences of 9's. Every positive real number has a unique decimal representation $a_0.d_1d_2\ldots$ where a_0 and $(d_i)_{i=1}^{\infty}$ are as in the above theorem.

4. Real numbers are uncountable

Theorem 9. *The set of real numbers is uncountable.*

Proof. It will suffice to show that there exists an infinite subset of the real numbers which is not countable. Let us consider the set $S = \{x \in \mathbb{R} | 0 \leqslant x < 1\}$. This set is clearly not finite.The decimal representation of any number in S is of the form $0.d_1 d_2 \ldots$ where sequence $(d_i)_{i=1}^{\infty}$ satisfies the conditions in the statement of Theorem .

Suppose S is a countable set. Then, there exists a bijection $x : \mathbb{N} \to S$, and we write the element $x(n)$ as x_n. In other words, all the elements of S occur in the sequence $(x_1, x_2, \ldots)$.

Let the decimal representation of x_i be $0.a_{i1} a_{i2} \ldots$. Now, we construct a new sequence $(c_i)_{i=1}^{\infty}$ as follows:

$$c_i = \begin{cases} a_{ii} + 1 & \text{if } a_{ii} < 7 \\ a_{ii} - 1 & \text{if } a_{ii} \geqslant 8. \end{cases}$$

Then, $(c_i)_{i=1}^{\infty}$ is a sequence that satisfies the conditions in Theorem 4. However, this sequence is necessarily different from the sequence $(a_{ji})_{i=1}^{\infty}$ for any natural number j. Thus, the real number x having decimal representation $0.c_1 c_2 \ldots$ is not equal to x_j for any j. However, we clearly have $x \in S$. This is a contradiction.

Thus, we conclude that S is not a countable set. □

5. Reading suggestions

The statement that the set of real numbers is uncountable is proved in Section 1.6 of the text ([1]). However, the detailed discussion regarding the existence and uniqueness of decimal representations is not given in the text. It has been briefly mentioned in Exercise 1.5.7.

References

[1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. 5